Optimal Control, Lecture 1: Introduction and Shortest Path

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Contents

1. Formulation of optimal control problems
2. Dynamic programming
3. Model predictive control
4. Reinforcement learning
5. Pontryagin maximum principle
6. Numerical methods

Applications



Applications



Applications

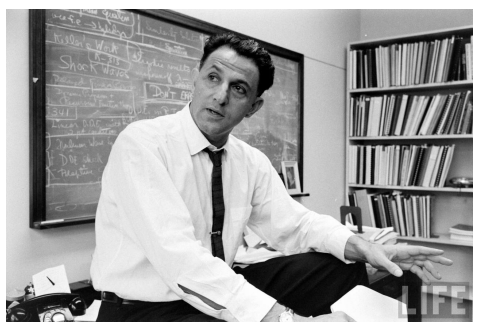


Applications

- Routing of airplanes
- Logistics
- Economical systems
- many more

History

- Roots in calculus of variations (Bernoulli, Euler, Lagrange, Weirstrass,...)
- Optimal control emerged in the 1950s during the space race
- Dynamic programming (Richard Bellman)
- Maximum principle (Lev Semenovich Pontryagin)
- Linear quadratic control (Rudolph Kalman)


Organization

- Examiner and lecturer: Anders Hansson, e-mail: anders.g.hansson@liu.se
- Course assistant: Reza Jafari, e-mail: seyyed.reza.jafari@liu.se (responsible for homeworks and exercise sessions)
- Course home page: http://www.control.isy.liu.se/student/tsrt08/

Course Evaluation and Course Development

- 22 registered students
- Evaliuate answer frequency 27.27%
- Overall evaluation of the course 3.17 (0x5, 2x4, 3x3, 1x2, 0x1)
- Criticism: Lectures stressed, more real-world examples, spend more time explaining when to use different approaches, in prerequisites only control mentioned, should also be math, linear algebra and optimization, exam questions different between different exams, hence need 5 hour exam, motivation lacking for what is done in the lectures, at the end when the overview was given it became more clear, lots of theory in the lectures, matrices within matrices confusing
- More motivating examples introduced.
- Praise: I found the lessons to be very helpful. The book , the lessons and the lectures. Lessons where good and both the examiner and the assistant was very helpful if you had questions.

Optimal Control

- Discrete time optimal control
- Continuous time optimal control

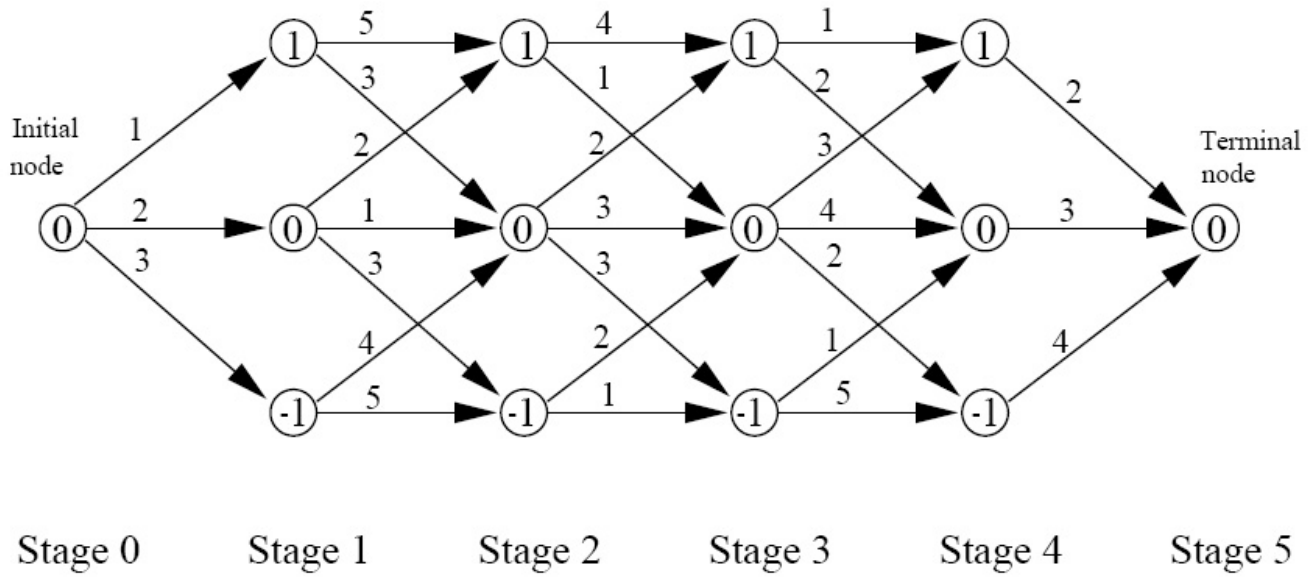
Discrete Time Optimal Control

$$
\begin{array}{ll}
\text{minimize} & \phi(x_N) + \sum_{k=0}^{N-1} f_k(x_k, u_k) \\
\text{subject to} & x_{k+1} = F_k(, x_k, u_k) \\
& x_0 \text{ given } , x_k \in X_k \\
& u_k \in U(k, x_k)
\end{array}
$$

Sequential decision problem which include as special cases

- Graph search problem
- Combinatorial optimization
- Discrete time optimal control

Shortest Path Problem



Find the shortest path from the initial node at stage 0 to the terminal node at stage 5.

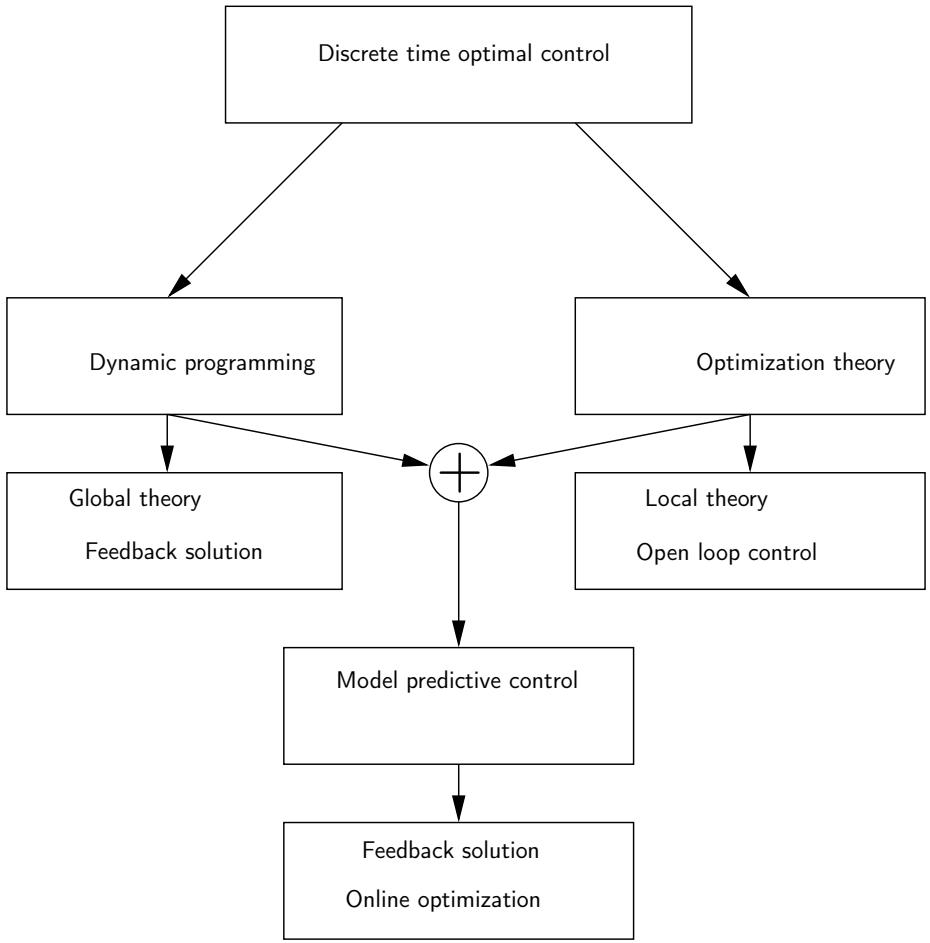
Knapsack Problem



maximize $\sum_{j=1}^{n} p_j x_j$

subject to $\sum_{j=1}^{n} w_j x_j \leq c, x_j = 0 \text{ or } 1, \ j = 1, \ldots, n$

Discrete Time Optimal Control



Reinforcement Learning

Is about solving the optimal control problem *without knowing* the dynamical model of the system to control.

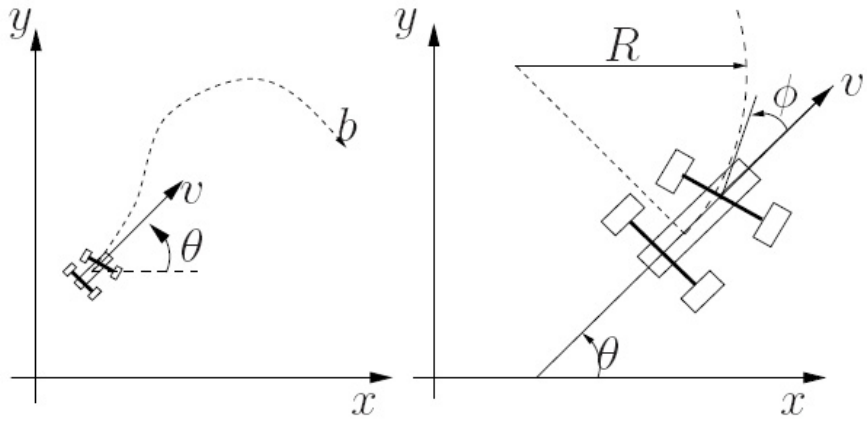
Based on carrying out experiments.

Continuous Time Optimal Control

$$
\begin{array}{ll}
\text{minimize} & \phi(t_1, x(t_1)) + \int_{t_0}^{t_1} f(t, x(t), u(t))dt \\
\text{subject to} & \dot{x}(t) = F(t, x(t), u(t)) \\
& x(t_0) \in S_0 \\
& x(t_1) \in S_1 \\
& u(t) \in U(x(t))
\end{array}
$$

S_0 and S_1 manifolds (intersections of surfaces)

Optimal Control of Car



Problem: shortest path (here t_1 is a free variable)

$$
\begin{array}{ll}
\text{minimize} & \int_0^{t_1} 1 dt \\
\text{subject to} & \dot{x}(t) = v\cos(\theta(t)),\ x(0) = 0,\ x(t_1) = \bar{x} \\
& \dot{y}(t) = v\sin(\theta(t)),\ y(0) = 0,\ y(t_1) = \bar{y} \\
& \dot{\theta}(t) = \omega(t),\ \theta(0) = 0,\ \theta(t_1) = \bar{\theta} \\
& |\omega(t)| \leq v/R,\ t_1 \geq 0
\end{array}
$$

Resource Allocation

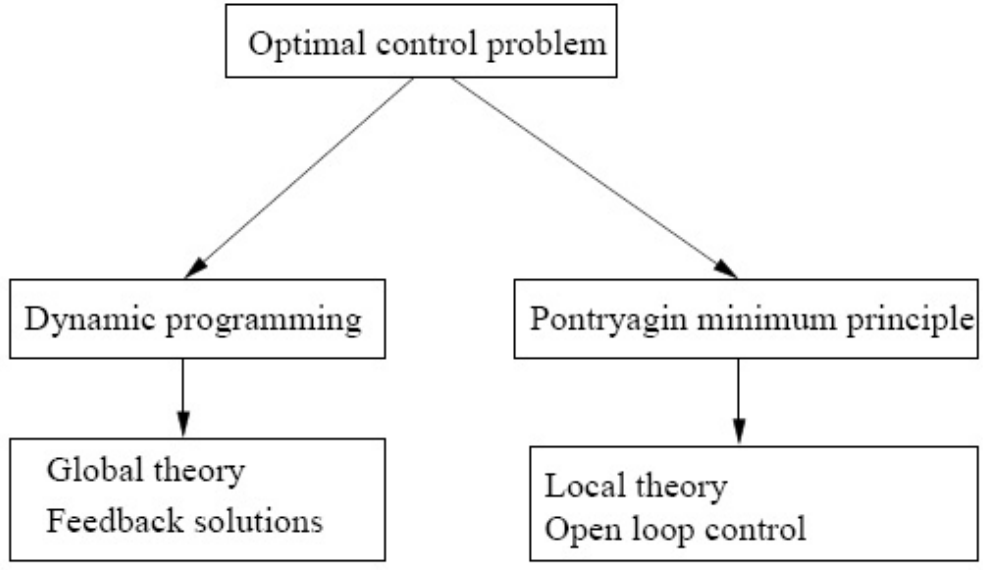
$$
\begin{array}{ll}
\text{maximize} & x(t_1) + \int_0^{t_1} (1 - u(t))x(t)dt \\
\text{subject to} & \dot{x}(t) = \alpha u(t)x(t),\ 0 < \alpha < 1 \\
& x(0) = x_0 > 0 \\
& u(t) \in [0, 1],\ \forall t
\end{array}
$$

The portion $u(t)$ of the production rate $x(t)$ is invested in the factory (and increases the production capacity)

The rest $(1 - u(t))x(t)$ is stored in a warehouse.

Maximize the sum of the total amount of goods stored and the final capacity.

Continuous Optimal Control



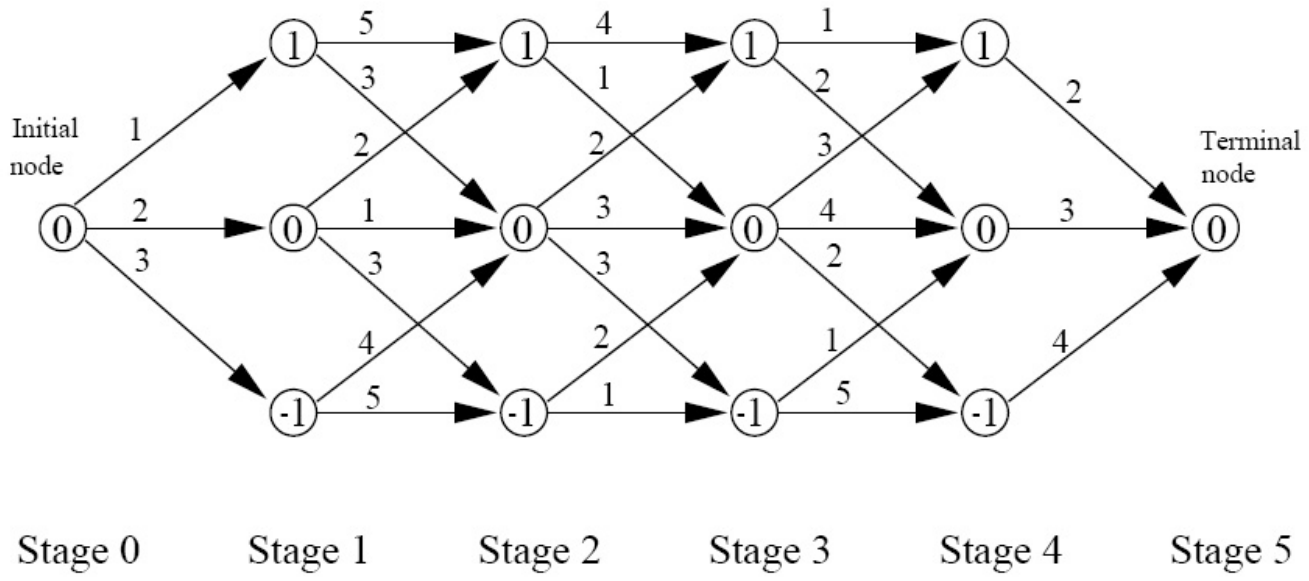
- Feedback control $u(t) = \psi(t, x(t))$ (depends on state)
- Open loop control $u(t) = \psi(t, x(t_0))$

Note: we will not do dynamic programming in continuous time.

Discrete Time Optimal Control

- Optimal control problems in discrete time
  - Discrete state space
  - Continuous state space
- In discrete time it is easier to understand
  - The principle of optimality
  - The dynamic programming equation
- Discrete version of the Pontryagin Maximum Principle (PMP)
- Optimal control on infinite time horizon
- Feedback versus open loop control
- Approximate solutions
- Model Predictive Control (MPC)
- Reinforcement learning

Shortest Path Problem



Find the shortest path from the initial node at stage 0 to the terminal node at stage 5.